				[2]		
8	**(a)**		$\begin{pmatrix}6\\8\end{pmatrix}-\begin{pmatrix}-3\\20\end{pmatrix}$	**M1**	May see $\begin{pmatrix}9\\-12\end{pmatrix}$ or $\begin{pmatrix}-9\\12\end{pmatrix}$	
			$\lvert\overrightarrow{AB}\rvert = 15$ or $\sqrt{225}$ so $\lvert\overrightarrow{AB}\rvert < 200$ or $\overrightarrow{\lvert AB\rvert}$ is not greater than 200	**A1**	May see $\sqrt{9^2+(-12)^2}$ oe CWO	
				[2]		
8	**(b)**		$\overrightarrow{AC} = \begin{pmatrix}18\\-24\end{pmatrix} = 2\times\begin{pmatrix}9\\-12\end{pmatrix}$	**M1**	FT their AB; allow mark for comparison of their AB and AC	
			so A, B and C are collinear	**A1**		
				[2]		